Siddhartha Sen

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Profile Summary

Final year Master's Data Science student with hands-on experience in data analysis and machine learning. Worked on dashboards, data preparation pipelines, healthcare analytics and network analysis.

Internships

Data Science Intern, Onlei Technologies, Noida Pvt. Ltd. May 2023 - October 2023

Automated Web Scraping & NLP-based Text Analytics (Python)

Designed and implemented a pipeline to scrape and process articles from multiple web URLs using Python, Requests and BeautifulSoup. Developed an NLP-based text analytics system using NLTK to compute 13 linguistic and readability metrics. Used VaderSentiment to categorise sentiments as positive, negative and neutral.

Advanced Analytics Intern, PharmaQuant Insights Pvt. Ltd. May 2025 - June 2025

Conducted meta-analysis across 7 clinical trials evaluating treatments for platinum-resistant ovarian cancer. Synthesized pooled efficacy across 700+ aggregated patient outcomes, measuring key endpoints using fixed & random-effects models. Identified high heterogeneity ($I^2 > 90\%$) among trials. Built forest plots and summary artefacts, enabling faster decision-support reporting.

Technical Skills

Programming	Python, R, SQL
ML, DL	Scikit-learn, XGBoost, Random Forest, Regression, Classification, Feature Engineering, CNN, LSTM, NLP
Data Handling	Pandas, NumPy
Visualization	Power BI, Matplotlib, Seaborn, Excel

Education

2024 - 2026	M.Sc. (Data Science) at **St. Xavier's College (Autonomous), Kolkata** (CGPA 7.89 till sem 3)
2020 - 2022	M.Sc. (Physics) at **St. Xavier's College (Autonomous), Kolkata** (Percentage : 84.2)
2017-2020	B.Sc. (Physics) at **St. Xavier's College (Autonomous), Kolkata** (Percentage : 73.28)
2017	Class 12th ISC Board (Percentage : 95.5 with 100 in Mathematics)

Major Projects

Predicting Avoidable Readmissions in U.S. Hospitals facing re-admission penalties (Diabetes Dataset).
ML models used - Logistic Regression, Random Forest, XGBoost.

- Built readmission-risk model isolating top driver features (treatment gaps, comorbidities, medication-timeline) explaining $\sim$ 65–70% of avoidable readmissions.
- Delivered hospital-focused business recommendations through interactive dashboards; positioned to reduce HRRP-penalty exposure to US based hospitals by improving care-coordination workflows.

Stock Price Prediction, Weight Optimization – Using Scipy CG, Custom Conjugate Gradient & normal equation.

- Built a quantitative finance engine via a reusable StockPriceOptimizer class integrating analytical and iterative solvers; reduced optimization convergence time by $\sim$ 35–40% compared to steepest descent.
- Achieved $\sim$ 10–12% improvement in forecasting accuracy by optimizing step size using Fibonacci line search within a custom conjugate gradient framework compared to steepest descent.

Identifying influential members in a social network and hiding them while maintaining indirect influence.

- Identified influential member using centrality measures. Engineered new adversarial rewiring strategies (C3R, Chameleon heuristics) enabling a targeted "mastermind" node to drop in centrality rankings after few iterations, proving structural vulnerability in degree, closeness & betweenness-based detection systems.
- Used diffusion based Independent Cascade (IC) and Linear Threshold (LT) metric to measure influence preservation. The IC and LT metric trends matched with theoretical expectations, at par with state of art algorithms.

Achievements and other activities

- **WINNER of PowerQuest Hackathon 2025 (HEOR Statistics)**, PharmaQuant Insights Pvt.Ltd.
- Completed and obtained badge for "Unleashing the Power of AI Agents" on IBM SkillsBuild.
- Google Advanced Data Analytics Credential Course(Ongoing).